

Highlights

- Factorial study decomposes framing into text, structure, role, and acoustic style.
- Narrative structure raises attack success by 27.9 pp (OR 16.65).
- Paired cross-modal safety divergence is model-dependent, not modality-intrinsic.
- MSRF is a knowledge-driven fusion of four signals (225 params; ROC-AUC 0.9784).
- Honest reporting: defense gate G2 fails; fixed thresholds do not transfer.

Cross-Modal Safety Inconsistency in Large Audio-Language Models: Measurement, Mechanism, and Knowledge-Driven Multi-Source Fusion

Yongjin Chen^a, Songze Zhu^{a,1}

^a*People’s Public Security University of China, China*

Abstract

Large audio-language models (LALMs) are deployed in safety-sensitive decision contexts, yet the safety implications of *adversarial framing* — repackaging a fixed request through narrative, role-play, or acoustic style — remain poorly understood. We treat this as a *knowledge-system* problem and propose a knowledge-driven methodology: a safe decision must fuse heterogeneous knowledge sources — request intent, narrative structure, acoustic style, and scoring uncertainty — that current detectors represent inconsistently or not at all. In a controlled paired factorial experiment ($E_t \times N \times R \times A_s$, 24 cells; 16,200 response cells; three LALMs; two languages), narrative structure (N) raises attack success by 27.9 percentage points (95% CI [25.0, 30.9]; odds ratio 16.65 [14.10, 19.67]). The effect is partially—not wholly—length-mediated and robust across languages, models, and out-of-distribution anchors; the agreement-conditioned dual-judge estimate (38.0 pp) is disclosed as selection-inflated. We further show paired cross-modal safety divergence (PCSD) is model-dependent, so audio vulnerability is not intrinsic to the modality. We then build MSRF, a knowledge-driven multi-source fusion framework that fuses these signal families through calibrated out-of-fold scores in an auditable 225-parameter layer. It reaches 0.9784 ROC-AUC on the deployment frame (0.9455 ± 0.030 cross-validation), outperforming billion-parameter guards, with boundaries reported honestly (75.9% benign false-positive rate; adaptive-attack decay). Narrative structure exhibits a robust causal effect under controlled prompt interventions; representing that

*Corresponding author

Email address: 20196846@ppsuc.edu.cn (Songze Zhu)

effect as structured, multi-signal knowledge is a more durable foundation for LALM safety than either modal shortcuts or opaque detection.

Keywords: cross-modal consistency, knowledge-driven fusion, LALM safety, framing attack, causal attribution

1. Introduction

Large audio-language models (LALMs) now power voice assistants, multi-modal customer service, and hands-free interaction, and are increasingly deployed in safety-sensitive contexts. A growing body of work shows that such models can be steered toward harmful output by *adversarial framing*: keeping the underlying request fixed while changing how it is packaged through narrative, role-play, or acoustic style (Yu et al., 2026; Qian and Li, 2026; Li et al., 2026). These attacks are practical — they require no gradient access, operate over the channel the model was designed to consume, and are largely invisible to text-only safeguards.

Yet the mechanisms behind framing attacks remain poorly understood. Prior work treats “delivery” as a single coarse preset and compares whole attacks against whole defenses. It does not ask *which semantic component* — the emotional narrative text, the event-chain structure, the role assignment, or the acoustic style — drives attack success, nor whether these components interact. Moreover, causal attribution has remained at the micro (internal) layer where it is inaccessible to black-box practitioners, and existing audio safeguards are rudimentary relative to their text counterparts.

Obstacles to component-level analysis. Three obstacles have blocked component-level analysis. First, framing components are confounded with response length: narrative prompts elicit longer model outputs, and length itself correlates with harmful content, so a naive comparison cannot separate framing from verbosity (Section 6.1.4). Second, safety judgment is itself unreliable: single classifiers fail catastrophically on Chinese inputs (FNR ≥ 0.96 for three off-the-shelf scorers) and disagree with each other on roughly one in six responses, so any measurement must rest on an audited multi-scorer protocol (Section 6.4.4). It also means a reliable safety decision cannot trust any single source and must fuse heterogeneous, mutually non-overlapping signals (Section 5). Third, large audio-language models form a heterogeneous family — a finding that holds for one architecture may not hold for another (Section 6.2).

32 *Connection to Knowledge-Based Systems..* We cast the problem as a
 33 *knowledge-system* problem rather than only a safety one. The four signals
 34 our study manipulates or measures — request intent, narrative structure,
 35 acoustic style, and scoring uncertainty — are four heterogeneous knowledge
 36 sources about a single decision (is this interaction harmful?). MSRF is
 37 a knowledge-fusion method that represents each source’s evidence, cali-
 38 brates it, and combines it under uncertainty into an auditable score. The
 39 paired cross-modal divergence (PCSD) quantifies a *cross-modal knowledge*
 40 *inconsistency* — the same request elicits different safety-relevant knowledge
 41 through text versus audio. This aligns the work with the knowledge-systems
 42 tradition of heterogeneous knowledge fusion, uncertainty reasoning, and
 43 decision support (Guo et al., 2024; Shao et al., 2024; Xu and Li, 2025; Zhao
 44 et al., 2025; Shafer, 1976) and with recent KBS contributions that treat
 45 safety as a knowledge- representation and decision-support problem (Shi
 46 et al., 2026; Yi et al., 2024).

47 *Contributions..* We report a controlled, paired factorial study that de-
 48 composes the four framing components across text and audio channels,
 49 a response-level cross-modal consistency metric, and a signal-level fusion
 50 framework — treating attack mechanisms as structured, auditable, multi-
 51 signal knowledge rather than opaque activations, an orientation shared with
 52 knowledge-based, decision-support systems (Shi et al., 2026; Yi et al., 2024).
 53 Concretely:

- 54 (i) **A knowledge-driven multi-source fusion framework.** We build
 55 Multi-Source Risk Fusion (MSRF), which represents four mutually
 56 non-overlapping signal families (harmful-intent semantics, narrative
 57 structure, acoustic prosody, and scoring uncertainty) as explicit
 58 knowledge sources and fuses them through out-of-fold calibration
 59 and a 225-parameter, auditable fusion layer. On the deployment
 60 frame it reaches 0.9784 ROC-AUC (0.9455 ± 0.030 over five frames),
 61 outperforming ShieldGemma, WildGuard, and a reproduced GradSafe
 62 (ROC-AUC 0.198, 0.174, and 0.315, respectively) on the same test
 63 frame, while using roughly two–three orders of magnitude fewer
 64 parameters. We report its limits honestly: fixed-threshold detection
 65 does not transfer to benign inputs (75.9% benign false-positive rate
 66 at the calibrated threshold), and adaptive attacks degrade detection
 67 (up to 20.0 pp). We therefore present MSRF as a *discriminative*,

68 *in-distribution* framework for mechanism-aware risk scoring, not as a
 69 turnkey deployment defense.

70 (ii) **Full-factorial causal attribution.** We manipulate E_t (narrative text
 71 framing), N (narrative structure / event-chain), R (role assignment),
 72 and A_s (acoustic style) in a 24-cell paired design over three large audio-
 73 language models, two languages, and an out-of-distribution anchor set.
 74 Narrative structure is the dominant causal driver: on the full sample,
 75 attack success rises from 2.2% to 30.1% ($\Delta 27.9$ pp [25.0, 30.9]; odds
 76 ratio 16.65 [14.10, 19.67]), with a consistent effect across languages (zh
 77 29.3 pp / en 26.6 pp), across models, and on the out-of-distribution Ad-
 78 vBench anchor ($\Delta 38.0$ pp, dual-judge robustness caliber). A response-
 79 length covariate mediates a substantial share of the raw effect, sup-
 80 porting an *elaboration-channel* account rather than a modality-intrinsic
 81 one. Role assignment has a *negative* effect (odds ratio 0.53) — a result
 82 we replicate and interpret rather than smooth over.

83 (iii) **Response-level cross-modal consistency.** We quantify paired
 84 cross-modal safety divergence (PCSD) as a paired consistency score
 85 over the safety-relevant properties of the text and audio rendering
 86 of the same input, and show it is model-dependent (e.g., near-zero
 87 divergence for one architecture, strongly positive for two others) —
 88 evidence that audio “vulnerability” is not intrinsic to the modality.
 89 We report it as an auxiliary diagnostic rather than a standalone
 90 contribution.

91 We pre-registered two decision gates — G1 (attribution established) and
 92 G2 (defense value); G1 passes, G2 does not, and we report both outcomes
 93 rather than only the defense-positive one. The remainder of this paper
 94 presents the protocol and factorial design (Section 3), the knowledge-driven
 95 fusion framework and its formalization (Section 5), the attribution results
 96 (Section 6.1), the cross-modal analysis (Section 6.2), the fusion detection re-
 97 sults and their honest boundaries (Section 6.3), adaptive and degradation
 98 robustness (Section 6.4), and discussion (Section 7), followed by limitations
 99 (Section 8) and conclusion (Section 9). Table 1 positions this work against
 100 the most related prior work across the four dimensions we contribute.

Table 1: **Positioning against the most related prior work.** Compared across the four dimensions of this paper’s contribution. ✓ = present; ◦ = partial (a single framing factor isolated without full decomposition); (✓) = a benchmark-level aggregate rather than a response-level measurement. All attributes follow the characterizations in Section 2.

Work	Component-level attribution	Cross-modal consistency	Fusion-based defense	Honest boundary reporting
Yu et al. (2026)	–	–	–	–
Qian and Li (2026)	◦	–	–	–
Li et al. (2026)	–	–	–	–
Pan et al. (2025)	–	(✓)	–	–
Xu et al. (2024)	–	✓	–	–
This work	✓	✓	✓	✓

2. Related Work

2.1. Framing and delivery attacks against LALMs

Audio-side attacks on large audio-language models have recently been shown to succeed via narrative delivery (Yu et al., 2026) and prosody (Qian and Li, 2026), and style-aware audio jailbreaks (Li et al., 2026) demonstrate that acoustic style alone can steer output. Paralinguistic variation, such as speaker emotion, similarly induces safety inconsistencies (Feng et al., 2026b), and a recent taxonomy organizes audio jailbreaks into semantic, acoustic, signal, and embedding layers, with narrative framing as a semantic-layer threat (Feng et al., 2026a). These works establish that framing is a real attack surface, but treat delivery as an atomic preset; none decomposes the semantic components or measures their causal contributions under control (Qian and Li, 2026; Yu et al., 2026; Li et al., 2026; Feng et al., 2026b,a). Our design differs on exactly this axis: where Yu et al. (2026) and Li et al. (2026) compare whole attacks against whole defenses, we decompose the “storytelling” bundle into four orthogonal components (text, structure, role, acoustic style) and attribute success to each under a paired control. Where Qian and Li (2026) varies prosody in isolation, our acoustic factor is only one of four, and we disclose that it confounds speaker identity and gender rather than claiming it isolates prosody (Section 4). Text-side research has taxonomized such persuasion strategies (Zeng et al., 2024b); our factorial design is the component-level complement that quantifies their causal contribution in the audio-language setting. The increment over this line of work is the decomposition itself: rather than proposing a new delivery variant, we provide a

125 reusable, controlled measurement of *which* framing component carries the
126 effect, under a paired design that separates structure from text, role, and
127 acoustic style.

128 2.2. Safety benchmarks and consistency measurement

129 Benchmarks such as HarmBench (Mazeika et al., 2024), Audio Jailbreak
130 (Song et al., 2025), and Omni-SafetyBench (Pan et al., 2025) evaluate
131 whether models or guards are safe on static collections, and Omni-
132 SafetyBench further defines a benchmark-level cross-modal consistency score
133 (CMSC). Consistency at the benchmark level, however, aggregates outcomes
134 and cannot localize per-input divergence between the text and audio
135 rendering of the *same* query — precisely where deployment disagreements
136 originate. We therefore define PCSD at the response level as a measurement,
137 not a benchmark, and operationalize the distinction in Section 6.2 (Pan
138 et al., 2025). The increment is response-level granularity: PCSD scores the
139 text and audio rendering of the *same* input per cell, which a benchmark-level
140 aggregate cannot localize, and it is reported as an auxiliary diagnostic signal
141 rather than a label.

142 2.3. Jailbreak detection and defense

143 Jailbreak behavior in aligned models has been traced to brittle safety-
144 training boundaries (Wei et al., 2023), and detection approaches span gradi-
145 ent analysis (Xie et al., 2024a), input-output safeguards (Inan et al., 2023;
146 Zeng et al., 2024a; Han et al., 2024), cross-modality consistency checking
147 for multimodal LLMs (Xu et al., 2024), confidence- and codebook-based
148 detection (Chen et al., 2025; Alanova et al., 2026), and audio authentic-
149 ity verification (Xie et al., 2024b). These detectors share a single-cue de-
150 sign: consistency disagreement (Xu et al., 2024) or first-token confidence
151 (Chen et al., 2025; Alanova et al., 2026) alone, with no explicit packaging-
152 structure signal. Our fusion framework instead combines four mutually non-
153 overlapping, interpretable signal families and, unlike prior work, is evaluated
154 on both in-distribution discrimination and honest deployment boundaries
155 (Section 6.3.4). On the defense side, inference-time refusal steering for large
156 audio-language models has also been proposed (Lin et al., 2026). Within the
157 *Knowledge-Based Systems* scope, jailbreak mitigation has been framed as a
158 model-agnostic semantic problem (Shi et al., 2026) and safety as a subspace-
159 oriented model-fusion objective (Yi et al., 2024); the journal’s orientation

160 toward knowledge representation and interpretable, decision-support detec-
 161 tion is our target (Shi et al., 2026; Yi et al., 2024). Relative to this literature,
 162 our contribution is three-fold: (a) causal attribution of *which* framing com-
 163 ponent matters (a knowledge question, not a classifier question); (b) a fusion
 164 framework whose branches are mutually non-overlapping, interpretable sig-
 165 nal families rather than opaque embeddings; and (c) an honest evaluation
 166 that separates in-distribution discrimination from deployment-level threshold
 167 behavior, which most detector papers do not report.

168 2.4. Knowledge fusion and decision-support systems

169 Within the knowledge-systems tradition, safety-relevant and operational
 170 decisions are supported by fusing heterogeneous knowledge sources under an
 171 explicit treatment of uncertainty: belief-theoretic uncertainty reasoning and
 172 quantification (Guo et al., 2024), evidential (Dempster–Shafer) fusion of mul-
 173 timodal information for reliable decisions (Shao et al., 2024), multi-criteria
 174 fusion over multi-source decision data (Xu and Li, 2025), and ensemble deci-
 175 sions over multimodal decision-information systems under uncertainty (Zhao
 176 et al., 2025). These works share a methodology that this paper adopts: rep-
 177 resent each source’s evidence explicitly, calibrate or discount it, and combine
 178 it under uncertainty to support a decision — the same structure that MSRF
 179 instantiates for LALM safety scoring with four mutually non-overlapping
 180 signal families (Shafer, 1976). Our increment over this line is twofold: the
 181 knowledge sources are *response-level signals extracted under a pre-registered*
 182 *causal protocol* rather than generic features, and the fused decision is eval-
 183 uated both for in-distribution discrimination and for its honest boundary
 184 behavior (fixed-threshold transfer), which decision-fusion studies typically
 185 do not report.

186 3. Protocol and Factorial Design

187 Full machine-readable protocol: RESEARCH_PROTOCOL.md v6.5 and
 188 STAGE_CONTRACTS.md; this section is the paper-facing summary.

189 3.1. Research questions and gates

190 **RQ1 (attribution).** Which framing component(s) causally increase attack
 191 success in large audio-language models, and are effects robust across
 192 language, model, and out-of-distribution inputs?

Table 2: **The four framing factors.** Varied in the paired factorial design (§4).

Factor	Definition	Levels
E_t	Narrative text framing (emotional / narrative wording)	neutral vs. narrative
N	Narrative structure (event-chain packaging)	direct request vs. chained narrative
R	Role assignment (identity + task)	none vs. expert role
A_s	Acoustic style (TTS rendering)	neutral voice vs. styled voice

193 **RQ2 (cross-modal).** Do text and audio renderings of the same input di-
 194 verge in safety judgment, and is the divergence model-dependent?

195 **RQ3 (detection).** Can a structured, signal-level fusion framework discrim-
 196 inate framed attacks in-distribution, and how does it behave under
 197 adaptive attack and audio degradation?

198 We pre-registered two decision gates. **G1** (attribution established) re-
 199 quires a consistent, ≥ 10 pp effect whose bootstrap CI excludes zero under
 200 the dual-judge caliber with acceptable scoring dispute. **G2** (defense value)
 201 requires a significant TPR gain, ≥ 3 contributing branches, and a benign
 202 false-positive rate $\leq 5\%$. As reported in Sections 6 and 7, G1 passes and **G2**
 203 **does not pass**; the paper therefore presents the attribution and mechanism
 204 results as primary and the fusion framework as an application-level validation
 205 with explicit boundary conditions.

206 3.2. Factors and manipulation

207 A paired factorial design (Fisher, 1935) varies four framing components
 208 while holding the harmful request fixed:

209 Three templates instantiate each condition (9 templates total); a ma-
 210 nipulation check on response-level markers confirms each textual factor was
 211 delivered (mean marker deltas 0.37–0.60, threshold 0.95 keep rate 1.0). We
 212 disclose that the acoustic factor is implemented via TTS voices (a female
 213 neutral voice vs. a male styled voice), so it confounds speaker identity and
 214 gender with prosody. We do not claim A_s isolates prosody, and we verify no

215 systematic length difference across A_s variants (mean difference 1.18 chars,
 216 $p = 0.24$).

217 4. Experimental Setup

218 4.1. Models and scorers

219 Three large audio-language models constitute the model matrix: the two
 220 primary Gemma-4-family models (E4B-it, audio+text; E2B-it, text) and
 221 Qwen2-Audio-7B as an architecture control that is excluded from primary-
 222 effect inference. All Gemma-family models run at BF16 with thinking dis-
 223 abled.

224 Safety judgment uses an audited multi-scorer protocol. Four scorers
 225 (HarmBench-clb (Mazeika et al., 2024), StrongREJECT (Souly et al., 2024),
 226 and two Gemma-family judges) are validated on an internal validation set
 227 (JailBench, $n = 602$)¹; the primary judge (judge_big) reaches 0.8555 ac-
 228 curacy. We define three calibers: *primary* (single best judge, full sample),
 229 *dual-judge* (two judges agree, else abstain), and *majority* (four-vote). We
 230 use the **primary** full-sample estimate for headline claims because it is the
 231 only caliber free of a selection bias we disclose next. We use the **dual-judge**
 232 caliber for cross-scorer robustness and for Chinese-language measurement,
 233 where single scorers fail catastrophically (HarmBench FNR = 1.0; Stron-
 234 gREJECT FNR = 0.99) while dual-judge reaches FNR 0.29 with 85.3% ac-
 235 curacy (the third off-the-shelf scorer, the Gemma-family judge, has FNR
 236 = 0.96). Because the dual-judge and majority calibers are defined only on
 237 rows where the two judges agree, and agreement correlates with condition
 238 (in the confirmatory experiment, storytelling rows are included at 57.9% vs.
 239 91.8% for baseline), these calibers systematically inflate the effect estimate
 240 ($\Delta 38.0$ vs. 27.9 pp). We report this agreement-conditioning selection bias
 241 in Section 6. We further disclose that all internal scorers are Gemma-family
 242 (three of the four majority votes; the four-vote majority double-counts the
 243 E2B family leg), and that agreement with an independent cross-family an-
 244 chor (Qwen2.5-32B-Instruct-AWQ) is high (0.929 on the E4B side; 0.962 on
 245 the full E2B text set), which bounds the same-family risk.

¹JailBench is not a public benchmark; it is reported here for reproducibility.

246 4.2. Statistical protocol

247 Effects are reported as paired per-query cluster bootstraps (Efron and
 248 Tibshirani, 1993) ($B = 10,000$) with 95% CIs, plus a Binomial Bayesian
 249 mixed-effects model (Bates et al., 2015) with query random intercepts and
 250 fixed model/template effects (the mixed-effect upgrade to full random struc-
 251 ture is not met — fewer than three models and three templates — and is
 252 reported as such; the benchmark-level cross-frame fusion table uses $B =$
 253 2,000). Paired per-query discordance is additionally tested with McNe-
 254 mar’s test (McNemar, 1947) on the baseline-safe/storytelling-harmful ver-
 255 sus baseline-harmful/storytelling-safe cells; the asymmetry is strong across
 256 languages and models (Chinese $\chi^2=34.6$, English $\chi^2=30.3$, E2B $\chi^2=83.2$,
 257 E4B $\chi^2=91.1$; all $p < 0.001$), and the same test is applied after 1:1 nearest-
 258 neighbor length matching (caliper 0.2 SD). We pre-registered TOST equiv-
 259 alence bounds (± 5 pp) (Lakens et al., 2018), O’Brien–Fleming sequential
 260 alpha spending, and manipulation-check thresholds. All generation is deter-
 261 ministic (byte-identical on re-run, 40/40), and decoding robustness is verified
 262 under sampling ($\Delta 20.7$ pp greedy vs. $\Delta 20.8$ pp sampled, $n = 48$).

263 5. A Knowledge-Driven Multi-Source Fusion Framework (MSRF)

264 Single-cue detectors are not trustworthy on framed inputs: off-the-shelf
 265 harmfulness classifiers fail catastrophically on Chinese inputs ($\text{FNR} \geq 0.96$;
 266 Section 4) and the strongest open guards score below chance on the framing
 267 distribution (Section 6.3). The preceding analysis also establishes that the
 268 effect is carried by a specific, identifiable component — narrative structure
 269 — not by generic “attackness” or modality. MSRF is a knowledge-driven
 270 response to both facts: it represents the evidence about a response as *four*
 271 *mutually non-overlapping knowledge sources*, calibrates each source’s score,
 272 and fuses them under an explicit uncertainty treatment into a single auditable
 273 score. Figure 1 summarizes the architecture.

274 5.1. Signal families as heterogeneous knowledge sources

275 Each signal family is a knowledge source $i \in \{\text{int}, \text{nar}, \text{ac}, \text{unc}\}$ that pro-
 276 duces a score $x_i(r) \in [0, 1]$ for a response r (its text or audio rendering of the

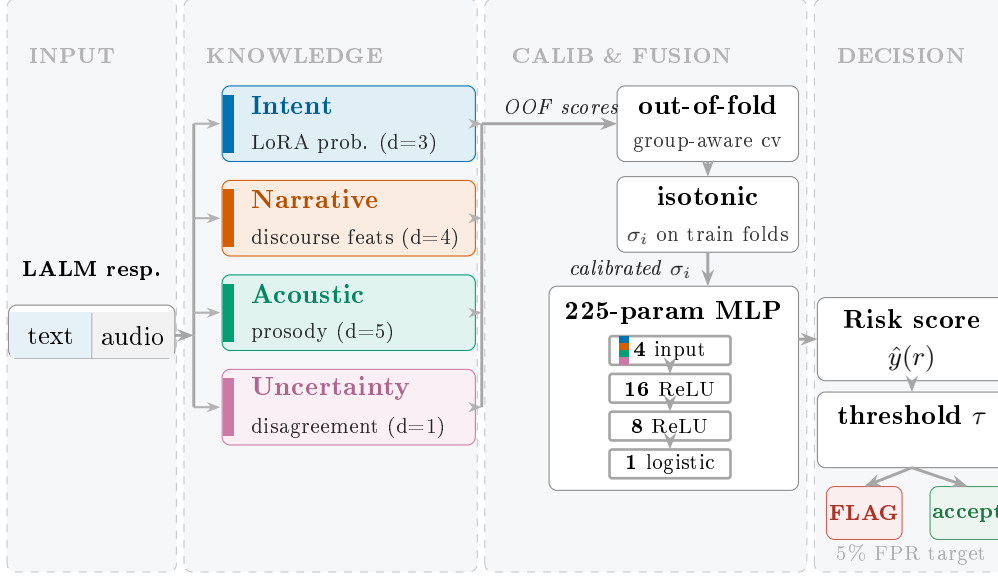


Figure 1: **The MSRF knowledge-driven fusion framework.** Four mutually non-overlapping knowledge sources — harmful-intent semantics (LoRA probability, $d=3$), narrative structure (discourse features, $d=4$), acoustic prosody ($d=5$), and scoring uncertainty (judge disagreement, $d=1$) — are extracted from each response, scored out-of-fold under group-aware splits (OOB scores), calibrated with isotonic regression (σ_i), and fused by a 225-parameter MLP ($4 \rightarrow 16 \rightarrow 8 \rightarrow 1$) into a discriminative risk score $\hat{y}(r)$; the operating threshold τ targets a 5% FPR operating point on the deployment frame. The honest boundary behavior of the framework (G2) is reported in Section 6.3.4.

277 same query):

$$278 \quad x_{\text{int}}(r) = g_{\text{int}}(p_{\text{int}}(r), p_{\text{int}}(r)^2, \ln(1 + \ell(r))), \quad (1)$$

$$279 \quad x_{\text{nar}}(r) = g_{\text{nar}}(\ell(r), n_{\text{sent}}(r), d_{\text{nar}}(r), n_{\text{story}}(r)), \quad (2)$$

$$280 \quad x_{\text{ac}}(r) = g_{\text{ac}}(f_0(r), e(r), \delta(r), \text{zcr}(r), r_{\text{sp}}(r)), \quad (3)$$

$$281 \quad x_{\text{unc}}(r) = 2 \cdot \min(p_{\text{unc}}(r), 1 - p_{\text{unc}}(r)). \quad (4)$$

282 The intent source x_{int} is a LoRA fine-tuned harmful-intent classifier over
 283 the E2B base whose input features are the intent probability, its square,
 284 and normalized response length ((1)); the narrative source x_{nar} uses in-
 285 interpretable discourse-structure features — length, sentence count, narrative
 286 density, and story-word presence ((2)); the acoustic source x_{ac} uses prosodic
 287 features (f_0 , energy, duration, zero-crossing rate, speech/pause ratios) with
 288 mean-imputation-plus-mask when audio is missing ((3)); and the uncertainty

source x_{unc} encodes scoring disagreement across the multi-scorer protocol as $2 \cdot \min(p, 1 - p)$, maximal when the scorer consensus is maximally uncertain ((4)). The families are designed to be non-overlapping: each covers a disjoint aspect of the response, so the fusion cannot “double count” the same evidence through two branches. We disclose two boundary conditions of the sources: the intent branch was trained in-distribution (its training rows overlap the evaluation set by design of the stacking protocol), and the acoustic branch has no measurable signal at our sample window ($N = 32$ attacks). We phrase the latter as “not measured”, not “ineffective”.

5.2. Fusion rule and parameter budget

Branch scores are calibrated with isotonic regression σ_i fit on training splits to render them comparable across families, then fused by a two-hidden-layer MLP $\mathcal{F}$:

$$\hat{y}(r) = \mathcal{F}\left(\sigma_1(x_{\text{int}}(r)), \sigma_2(x_{\text{nar}}(r)), \sigma_3(x_{\text{ac}}(r)), \sigma_4(x_{\text{unc}}(r))\right), \quad (5)$$

$$|\mathcal{F}| = (4 \cdot 16 + 16) + (16 \cdot 8 + 8) + (8 \cdot 1 + 1) = 225. \quad (6)$$

The fused layer is a $4 \rightarrow 16 \rightarrow 8 \rightarrow 1$ ReLU MLP with 225 trainable parameters ((6)); the operating threshold τ targets 5% FPR on the deployment frame. The parameter budget is a deliberate design choice: with 225 parameters the entire decision function is inspecting three numbers per source, so every contribution to the score can be traced back to an interpretable branch. This is an auditability property that multi-billion-parameter guards (ShieldGemma-9B, WildGuard, GradSafe) do not offer, and one we exploit to localize the honest boundary of the framework in Section 6.3.4. This is the knowledge-fusion position of Shafer (1976) and recent evidential fusion work (Guo et al., 2024; Shao et al., 2024; Xu and Li, 2025): evidence is made explicit and combined under uncertainty, rather than buried in a black box.

5.3. Evaluation protocol

Branch scores are produced *out-of-fold* so that no row contributes its own label to the score the fusion layer sees. The split is a group-aware Group-ShuffleSplit whose group key is template family $\times$ attack family (21 groups), so that an entire template $\times$ attack group is held out together and leakage across conditions is prevented. An outer 25% test frame never participates in hyperparameter selection, and an inner 20% validation split selects the MLP learning rate. The protocol is repeated over five independent seeds,

Table 3: **The 225-parameter fusion budget.** Layer-by-layer composition of the fusion MLP $\mathcal{F}$ ($4 \rightarrow 16 \rightarrow 8 \rightarrow 1$). The four inputs are the isotonic-calibrated branch scores $\sigma_1, \dots, \sigma_4$ (intent, narrative, acoustic, uncertainty); hidden activations are ReLU and the output is a logistic layer, matching the scikit-learn `MLPClassifier` used in the protocol (`hidden_layer_sizes=(16,8)`, `activation="relu"`).

Layer	Shape	Composition	Parameters
1	$4 \rightarrow 16$ (ReLU)	$4 \times 16 + 16$	80
2	$16 \rightarrow 8$ (ReLU)	$16 \times 8 + 8$	136
3	$8 \rightarrow 1$ (logistic)	$8 \times 1 + 1$	9
<i>Total trainable</i>			225

yielding five independent test frames with the deployment frame reported as primary. The silver ground-truth label is an agreement-based subset of the multi-scorer protocol (66.2% coverage; mean pairwise agreement 0.8122), used only to train and score the fused detector.

Algorithm 1 summarizes training and inference.

6. Results

Per the pre-registered protocol (Section 3), this section reports the confirmatory attribution test under the primary caliber first, then exploratory decompositions and diagnostic analyses; primary conclusions are anchored to the pre-registered main-effect hypotheses, and all other comparisons are reported as exploratory.

6.1. Attribution (RQ1)

6.1.1. Narrative structure is the dominant causal driver

In the confirmatory full experiment (P1-FULL; 16,200 cells = 600 queries $\times$ 3 conditions $\times$ 3 templates $\times$ 3 models, where the 600 queries are 400 in-distribution — 200 Chinese and 200 English — plus 200 out-of-distribution AdvBench anchors; text modality), storytelling framing raises attack success from **2.17%** to **30.08%** under the primary (judge_big) caliber on the full sample: Δ **27.9 pp** (95% CI [25.04, 30.92]; mixed-model **OR 16.65** [14.10, 19.67] with query random effects). The effect is robust in direction and significance across all three calibers; the consistency calibers yield larger magnitudes (dual-judge Δ 38.0 pp [33.98, 42.2]; majority Δ 36.8 pp) because they are defined only on rows where the two judges agree, and agreement correlates

Algorithm 1 MSRF training and inference

Require: responses R (text/audio renderings); silver labels $L \subset R$; groups G (template family $\times$ attack family)

Ensure: calibrated fusion model $\mathcal{F}$; threshold τ

```
1: procedure TRAIN( $R, L, G$ )
2:   for each family  $i \in \{\text{int, nar, ac, unc}\}$  do
3:     extract features and compute branch score  $x_i(r)$  for all  $r \in R$ 
4:     recompute branch scores out-of-fold over group-aware splits of  $G$ 
5:   end for
6:   fit isotonic calibrators  $\sigma_i$  on the training splits
7:   select MLP learning rate on the inner 20% validation split
8:   fit  $\mathcal{F} : \mathbb{R}^4 \rightarrow [0, 1]$  ( $4 \rightarrow 16 \rightarrow 8 \rightarrow 1$ ) on calibrated branch scores
9:   set  $\tau$  to target 5% FPR on the deployment-frame validation set
10:  return  $\mathcal{F}, \{\sigma_i\}, \tau$ 
11: end procedure
12: procedure INFER( $r$ )
13:  compute  $\{x_i(r)\}$ ; return  $\mathcal{F}(\sigma_1(x_1), \dots, \sigma_4(x_4))$ 
14: end procedure
```

346 with condition (storytelling inclusion 57.9% vs. baseline 91.8%), which sys-
347 tematically inflates these estimates. We disclose this agreement-conditioning
348 selection bias rather than smoothing it. The unrestricted-creativity condition
349 produces an intermediate effect ($\Delta 18.5$ pp on the dual robustness caliber),
350 and the out-of-distribution AdvBench anchor reproduces the storytelling ef-
351 fect under the same dual robustness caliber ($\Delta 38.0$ pp, computed on paired
352 dual-judge-eligible cells versus 30.8 pp pooled over the full sample; its odds
353 ratio of 363 is inflated by the near-zero baseline and is reported alongside the
354 percentage-point effect).

355 *6.1.2. Cross-lingual and cross-model robustness*

356 The effect is consistent across languages (Chinese $\Delta 29.3$ pp, English
357 $\Delta 26.6$ pp; interaction OR 1.41, $p < 0.001$, i.e., a modestly stronger ef-
358 fect in Chinese) and across the two primary models (E4B $\Delta 30.8$ pp, E2B
359 $\Delta 25.1$ pp) under the primary caliber; these per-language and per-model es-
360 timates average to ≈ 27.9 pp, consistent with the pooled headline. Under
361 the dual-judge robustness caliber, the benchmark main table reproduces the
362 same ordering across the model $\times$ language grid (range 29.9–40.2 pp), and the

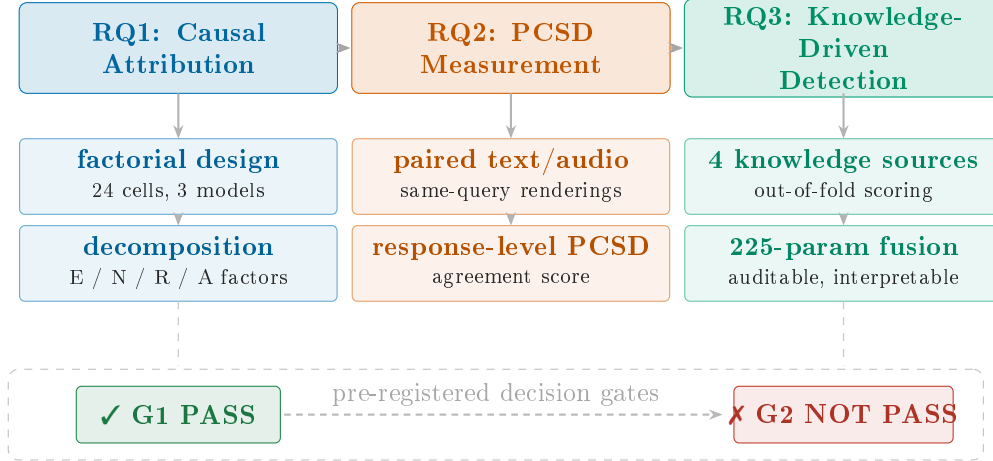


Figure 2: **Study overview: from causal attribution through cross-modal consistency to knowledge-driven detection.** RQ1 attributes attack success to narrative structure ($\Delta 27.9$ pp, OR 16.65); RQ2 quantifies response-level paired cross-modal divergence (PCSD) as model-dependent; RQ3 evaluates the knowledge-driven fusion detector MSRF (ROC-AUC 0.9784 on the deployment frame). The pre-registered gates are reported honestly: G1 passes and G2 does not (75.9% benign false-positive rate at the calibrated threshold; Section 6.3.4).

363 architecture-control model Qwen2-Audio-7B reproduces the direction on text
 364 (ASR 5.3% $\rightarrow$ 38.8%, $\Delta 33.5$ pp) while being excluded from the primary mixed
 365 model by design. Under our pre-registered cross-lingual consistency rule the
 366 effect is judged robust in direction and magnitude.

367 6.1.3. Benign control: framing amplifies existing intent

368 On an independent benign-input control ($n = 162$), storytelling framing
 369 produces essentially no harmful output (0.0% baseline, 3.2% storytelling,
 370 0.0% unrestricted; both flip directions zero). This supports an *amplification*
 371 account: framing does not conjure harmful behavior from benign inputs; it
 372 amplifies existing harmful intent.

373 6.1.4. The elaboration-channel hypothesis: length mediates a large share

374 The raw framing effect is confounded with response length: storytelling
 375 responses average 937 characters versus 417 for baseline. In the pilot factorial
 376 ($n = 14,400$ cells), including a response-length covariate collapses the N ef-
 377 fect in the mixed-effects model (OR 1.49 without length vs. 1.03 with length).
 378 On the full data, however, within-length-band analyses retain a substantive

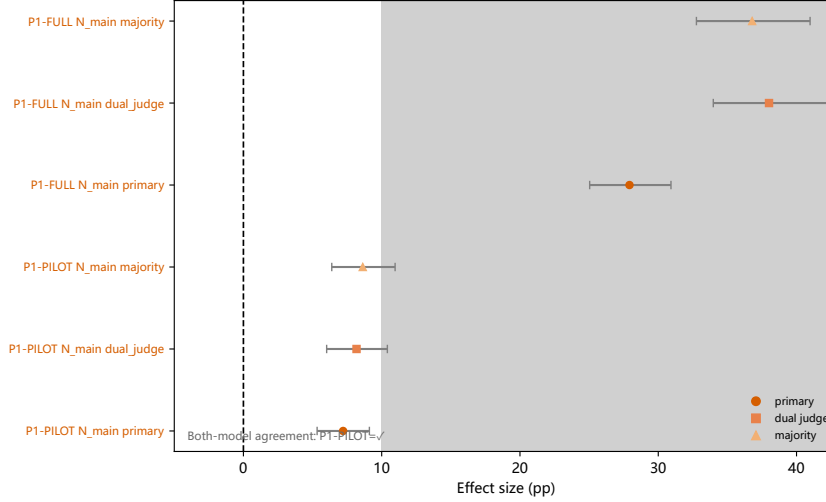


Figure 3: **Factorial forest plot.** Main and interaction effects of the framing components on attack success under the primary caliber, estimated from the pre-registered analysis; the zero line marks no effect, and the shaded band marks the practically meaningful region (≥ 10 pp). Intervals are cluster-bootstrap 95% CIs ($B = 10,000$); per-effect n as labeled.

379 content effect: in the only band with real baseline samples (200–799 chars)
 380 storytelling is not separable ($\Delta \approx +0.5$ pp), but in the ≥ 800 -char band —
 381 where 96% of storytelling responses fall — the storytelling effect is **+21.7 pp**
 382 (baseline 9.6% vs. storytelling 31.3%), and length-matching at ≥ 1012 chars
 383 still leaves **+17.5 pp**; the judge is not a “long $\Rightarrow$ harmful” shortcut, since
 384 baseline long responses score only 9.6% harmful. We therefore character-
 385 ize the effect as *partially—not wholly—length-mediated* (elaboration-channel:
 386 framing $\rightarrow$ longer elaboration $\rightarrow$ more harmful), with a real content effect
 387 surviving within matched length bands. On the audio channel the three esti-
 388 mators disagree in sign (single-slope OR 5.20; Mantel–Haenszel OR 0.60; 1:1
 389 matched -10.5 pp, n.s.), and length-matched comparisons render Qwen2-
 390 Audio’s audio-channel effect negative (-39.3 pp, $p = 0.003$). We report all
 391 three estimators rather than a single favorable one, and we do **not** claim that
 392 the audio modality is intrinsically more vulnerable.

393 6.1.5. Component decomposition: N drives, R protects

394 The factorial decomposition separates the components. Narrative struc-
 395 ture (N) carries the effect; narrative text framing (E_t) contributes a smaller
 396 but significant effect; **role assignment (R) has a negative effect** (OR

0.53, replicated across models) — assigning the model an expert identity makes harmful output *less* likely in our protocol, a counter-intuitive result we replicate and discuss in Section 7. Interactions are modest: $E_t \times R$ is significant in the pilot (OR 2.54), and the N effect is stronger on audio than on text (pilot dual-judge diff 7.26 pp; majority 9.84→22.66 on audio vs. 3.7→8.38 on text). Family-level stratification shows the effect generalizes across all five covered attack families (5/5 positive, 3/5 significant on dual-judge), with hate speech the most amplified ($\Delta \approx +17.6$ pp, roughly three times the overall effect; $N \times \text{hate}$ interaction significant in both models, +13.5 pp on E2B and +31.2 pp on E4B); coverage excludes network attacks and benign requests. Of 84 pre-registered comparisons, 65 are significant, all surviving BH-FDR correction (Benjamini and Hochberg, 1995) at $q = 0.10$.

6.2. Cross-modal consistency (RQ2)

We quantify **paired cross-modal safety divergence** (PCSD) through a per-input *agreement* score over the safety-relevant properties of a model’s response to the text rendering versus the audio rendering of the *same* query (length ratio and lexical overlap, combined 0.5/0.5; 0–1, higher agreement = less divergence). PCSD is a response-level measurement, distinct from benchmark-level consistency scores (Pan et al., 2025), and we use the complementary audio-minus-text attack-success gap Δ as a per-model divergence index.

Across the model matrix, PCSD is **model-dependent**: agreement is lowest for E4B (0.248) and highest for Qwen2-Audio (0.404), and the audio-minus-text gap mirrors this ordering (E4B +30.7 pp, E2B +44.6 pp, **Qwen2-Audio** −0.2 pp). Two implications follow. First, audio “vulnerability” is not intrinsic to the modality — one architecture shows no audio-text divergence at all. Input-side paralinguistic variation, such as speaker emotion, likewise shifts large audio-language model safety outcomes (Feng et al., 2026b), reinforcing that inconsistency is a property of the model–input interaction rather than of audio itself. Second, cross-modal safety inconsistency is a measurable, per-input phenomenon that a single modality-level metric would obscure; this motivates treating cross-modal divergence as an auxiliary diagnostic signal rather than a label.

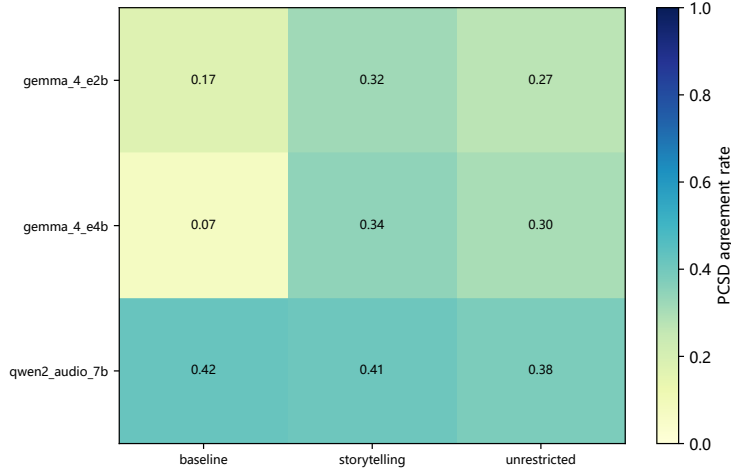


Figure 4: **PCSD paired-divergence heatmap**. Response-level agreement and divergence asymmetry between the text and audio rendering of the same query under attack conditions, across models and framing conditions; darker cells indicate higher agreement (less divergence). Per model $\times$ condition cell n as labeled.

6.3. A knowledge-driven fusion framework (RQ3)

6.3.1. Design

MSRF is presented as a self-contained method in Section 5: four mutually non-overlapping knowledge sources (harmful-intent semantics, narrative structure, acoustic prosody, scoring uncertainty), out-of-fold group-aware scoring, isotonic calibration, and a 225-parameter auditable fusion layer. We restate the headline protocol here for continuity and evaluate the framework below.

6.3.2. In-distribution discrimination

On the deployment frame ($n = 3,863$, 3.6% positive), MSRF reaches **ROC-AUC 0.9784**, PR-AUC 0.6119, TPR@FPR5 0.8345, ECE 0.0299. Over five independent group-split frames (protocol in Section 5; each frame is that seed’s group-aware test split, with the MLP learning rate selected on an inner 20% validation split so test rows never participate in selection), mean ROC-AUC is **0.9455 ± 0.030** (min 0.8909, max 0.9784). The weakest frame still outperforms the strongest open-source classifier by $2.8\times$ (Table 4).

Leave-one-branch-out ablation, evaluated relative to the five-frame mean AUC (0.9455), shows intent and narrative carry the signal (removing them

Table 4: **Cross-frame evaluation of the fused detector.** Five independent group-aware GroupShuffleSplit test frames (group key = template family $\times$ attack family, 21 groups; outer 25% test, inner 20% validation for learning-rate selection; Section 5). Frame 20260811 is the deployment frame used for all baseline comparisons.

Seed	Test n	ROC-AUC	PR-AUC	TPR@FPR5
20260811 (deployment)	3,863	0.9784	0.6119	0.8345
20260812	2,198	0.8909	0.6804	0.4608
20260813	3,323	0.9447	0.4750	0.6071
20260814	4,242	0.9667	0.7015	0.7891
20260815	3,149	0.9470	0.5555	0.5539
Mean over 5 frames	–	0.9455 ± 0.030	–	–

Table 5: **Leave-one-branch-out (LOO) ablation and single-branch performance.** Evaluated relative to the five-frame mean AUC (0.9455). The acoustic branch has no measurable signal at our sample window ($N = 32$ attacks; reported as “not measured”), and the uncertainty branch has no standalone score.

Branches	LOO ROC-AUC	Δ vs. full	Single-branch ROC-AUC
Full (all four)	0.9455	–	–
– intent	0.9136	–0.032	0.9325 ± 0.033
– narrative	0.9325	–0.013	0.9136 ± 0.048
– acoustic	0.9449	–0.001	0.5 ± 0.0 (no signal)
– uncertainty	0.9448	–0.001	–

448 drops AUC to 0.9136 and 0.9325, i.e., by 0.032 and 0.013) while the acoustic
449 and uncertainty branches contribute marginally (removal changes AUC by
450 ≤ 0.001 , to 0.9449 and 0.9448). Table 5 reports the full ablation alongside
451 each branch’s standalone score: the acoustic branch alone is chance-level (0.5,
452 no signal at our sample window), and the two best branches alone (intent
453 0.9325, narrative 0.9136) each fall short of the fused detector, so the fusion
454 gain is not attributable to any single knowledge source.

455 6.3.3. Comparison with open-source guards

456 On the same deployment frame, MSRF outperforms ShieldGemma-9B
457 (0.198 ROC-AUC), WildGuard (0.174), and a faithfully reproduced GradSafe
458 (0.315) while using 225 parameters versus multi-billion-parameter guards;
459 measured latency is 0.20 s / 6.7 GB for ShieldGemma and 1.04 s / 4.8 GB

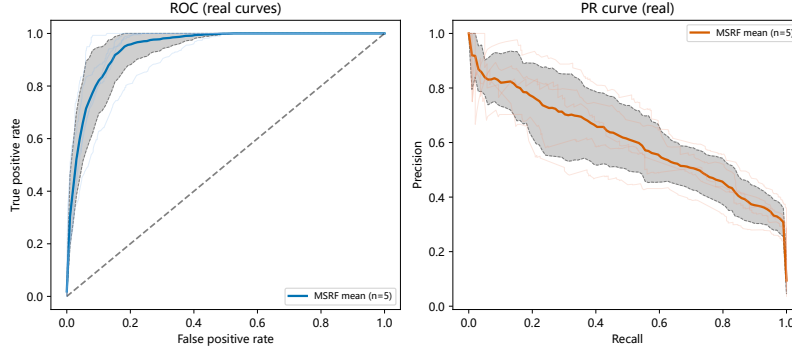


Figure 5: **MSRF ROC/PR curves.** Detection performance of the real fused detector on the deployment frame ($n = 3,863$, 3.6% positive). Solid curves are the 5-seed mean with a 10–90 percentile envelope (mean ROC-AUC 0.9455 ± 0.030 ; the deployment frame achieves 0.9784).

460 for WildGuard (Table 6). We note that all three open classifiers are *negatively*
 461 correlated with our safety labels on framed inputs (ROC-AUC < 0.5), which
 462 we interpret as systematic miss-classification of framing (flagging benign,
 463 missing harmful) rather than label inversion. Cross-frame replication (five
 464 independent frames) confirms MSRF wins every frame against the strongest
 465 baseline (margins 0.013–0.037).

466 6.3.4. Honest boundary: threshold behavior does not transfer

467 On an independent benign set ($n = 162$), the calibrated threshold pro-
 468 duces a **75.9% false-positive rate** for MSRF, versus 0.0% for ShieldGemma
 469 and WildGuard. The root cause is a distribution mismatch: the calibration
 470 pool consists of refusal responses, whereas true benign inputs yield compli-
 471 ant responses that score high under intent/narrative signals. We therefore
 472 present MSRF as a *discriminative, in-distribution* scoring framework — not
 473 a deployment threshold detector — and we report the threshold boundary
 474 explicitly. This is the criterion on which our pre-registered defense gate G2
 475 does not pass.

476 6.4. Robustness and adaptive attack

477 6.4.1. Defense decay on the attack set

478 On a separate attack set ($n = 2,864$, gray-box and white-box perturba-
 479 tions, each class $n \geq 200$), we re-ran the guards at the response level. On the

Table 6: **Open-source guard baselines.** Evaluated on the same deployment frame as MSRF ($n = 3,863$). ShieldGemma and WildGuard are real-inference runs; the GradSafe-real row is a reproduction of the ACL 2024 gradient-analysis method; the GradSafe proxy is a zero-additional-model gradient proxy in the same feature space. ROC-AUC below 0.5 indicates negative correlation with our safety labels on framed inputs (systematic misclassification: flagging benign, missing harmful), which we interpret rather than as label inversion.

Method	ROC-AUC	PR-AUC	Latency	Peak VRAM
MSRF (fusion, 225 params)	0.9784	0.6119	—	—
GradSafe proxy	0.9652	0.5961	—	—
ShieldGemma-9B	0.198	0.029	0.20 s	6.7 GB
WildGuard (Mistral-7B)	0.174	0.029	1.04 s	4.8 GB
GradSafe-real	0.315	0.024	—	—

harmful subset ($n = 52$), MSRF detects 15.4%, ShieldGemma 26.9%, WildGuard 48.1%; on all rows, MSRF 19.1%, ShieldGemma 11.3%, WildGuard 22.2%. We report these numbers in full: MSRF’s advantage is in-distribution discrimination (AUC), not fixed-threshold hit rate on attacks. We also disclose that the attack responses carry no LoRA intent signal (the intent branch sees an out-of-distribution zero), which explains part of the decay.

6.4.2. Adaptive attack decay

Against the real fused detector, gray-box and white-box perturbation classes show detection-rate decay up to 20.0 pp relative to baseline. We scope the claim precisely: our perturbations are single-pass, prompt-level transformations, not iterative or gradient-based optimization, so we claim *prompt-level perturbation robustness*, not adversarial robustness in the gradient sense. Three of six classes are effectively no-ops (fewer than half the perturbations change the input) and are reported as such rather than as attacks.

6.4.3. Audio degradation

Real audio degradation (Opus 16 kbps, SNR 20 dB, RIR reverberation, each EBU R128-normalized) changes attack success in opposite directions across models (e.g., -7.0 pp for E2B, $+11.3$ pp for E4B under Opus), which we report as-is. Degraded audio increases “thinking-trace leakage” in one model (10.0% / 26.0% / 28.5% for the three degradations vs. 7.3% baseline), and that model’s audio-side robustness must not be extrapolated to its text

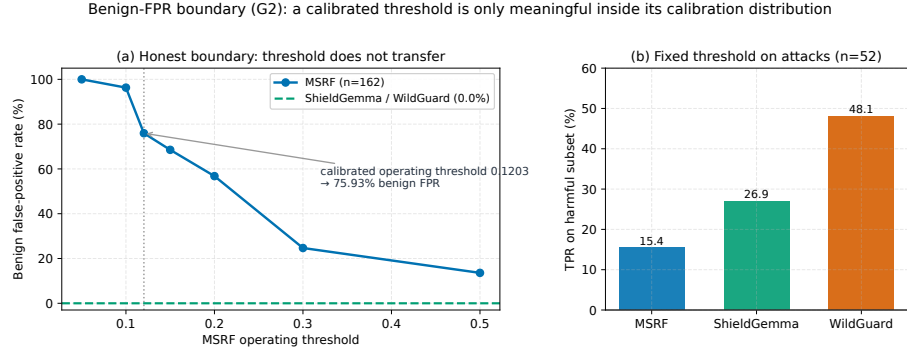


Figure 6: **Benign-FPR boundary (the G2 criterion)**. Rendered as evidence rather than avoided: (a) MSRF false-positive rate on the independent benign set ($n = 162$) rises to **75.9%** at the calibrated operating threshold 0.1203, versus 0.0% for ShieldGemma and WildGuard; (b) at that *fixed* threshold, TPR on the harmful subset ($n = 52$) is 15.4% for MSRF, 26.9% for ShieldGemma, and 48.1% for WildGuard — MSRF’s advantage is in-distribution discrimination (AUC), not fixed-threshold hit rate on attacks.

side, where empty refusals reach 58–85% (a genuine model behavior we report as such).

6.4.4. Measurement trust chain

We report a five-layer trust chain: deterministic generation (byte-identical, 40/40); scorer reliability (judge_big vs. judge_small agreement 0.82 on $n=3,573$ aligned cells, with Gwet’s AC1 0.745 and balanced κ 0.706 from the pipeline’s agreement utilities (Cohen, 1968)); cross-family scorer agreement (E4B side 0.929; full E2B text set 0.962); test–retest agreement (1.000 across three scorers); and Dawid–Skene latent-label recovery (0.945 in simulation at the measured error regime, 50 replications, 180 items, $P(\text{harm})=0.5$). Batch-inference byte-equality is not guaranteed (42/200 cells byte-identical, zero errors), which we disclose in the reproducibility statement.

7. Discussion

Why narrative structure, and why does role assignment protect? The factorial results separate the “storytelling” bundle into a strong structure effect (N) and a counter-intuitive protective effect (R). A post-hoc account consistent with both is the elaboration-channel hypothesis: N increases elaboration (length, narrative density), which increases both the probability and

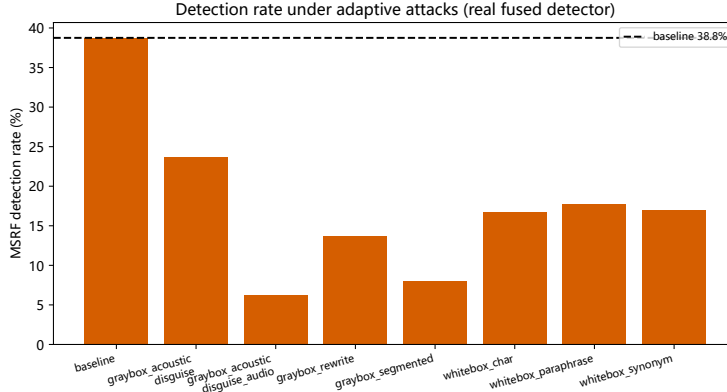


Figure 7: **Detection-rate decay under adaptive perturbations.** The real fused detector under gray-box and white-box adaptive perturbation classes, with decay up to 20.0 pp relative to baseline. Perturbations are single-pass, prompt-level transformations; we claim prompt-level perturbation robustness, not gradient adversarial robustness.

521 the perceived legitimacy of harmful continuations; R , by contrast, constrains
 522 the model into an identity whose stance (an “expert advisor”) is at odds with
 523 producing harmful instructions in our protocol. The replication of R ’s sign
 524 across models argues against a fluke, but its sensitivity to the scoring caliber
 525 (it reverses under latent-label scoring) demands caution; we report both.

526 *Why do open-source guards fail on framing?* ShieldGemma, WildGuard, and
 527 GradSafe all score below chance on framed inputs. This is consistent with
 528 their training objective (harmfulness of direct requests) being systematically
 529 mismatched with framing, which is safe-looking. It motivates a *knowledge*
 530 framing of the detection problem: the relevant signal is not “is this text
 531 harmful” but “does this structure package harmful intent,” which is exactly
 532 the signal our narrative and intent branches measure.

533 *Threshold detection versus mechanism-aware scoring.* The benign-FPR fail-
 534 ure (Section 6.3.4) is instructive rather than incidental. It shows that a cal-
 535 ibrated threshold is only meaningful inside the calibration distribution, and
 536 that compliant benign behavior is statistically confounded with framing sig-
 537 nals. Read as a *knowledge boundary* rather than a defect, this is a substantive
 538 empirical finding about the framework’s applicability domain: the calibration
 539 pool (refusal responses) and the benign pool (compliant responses) occupy
 540 different regions of the fused score space, so the two distributions are not

mutually calibrating. It is exactly this kind of domain knowledge that our pre-registered defense gate G2 was designed to elicit, and we report it in full. We argue that the honest contribution of a fusion framework in this setting is a *discriminative risk score with auditable branch attributions* — not a turnkey threshold — and we present the paper’s detection claims accordingly. A direct path forward is adaptive or online recalibration on deployment traffic, which we leave to future work. This aligns with the journal’s orientation toward interpretable, decision-support knowledge systems (Shi et al., 2026; Yi et al., 2024).

Generalizability of the methodology. The contribution is best read as a *methodology* rather than a tuned detector. The four knowledge sources — intent semantics, narrative structure, acoustic prosody, and scoring uncertainty — are defined from first principles about the response and the scoring process, not about a specific model or language; the causal protocol and the paired cross-modal measurement are likewise model-agnostic. We therefore expect the framework to transfer to other multimodal model audits (e.g., vision-language assistants whose safety responses also admit packaging structure), to other high-resource languages, and to high-stakes voice applications where an auditable, single-model risk score is preferable to a stack of opaque classifiers. What transfers is the *structure*: mutually non-overlapping, auditable knowledge sources, out-of-fold scoring, and honest boundary reporting. We have not re-run the full protocol on these settings, and we state this explicitly rather than claiming measured transfer.

Implications for knowledge-based decision systems. For the decision-support community, three design lessons follow. First, a safety decision over a model response is a multi-source fusion problem. The four signals we fuse are heterogeneous knowledge sources with different reliability (the intent and narrative branches carry the signal; the acoustic branch is “not measured”, not “ineffective”). Explicit calibration plus an auditable fusion layer lets a system operator trace which source drove a score. Second, the boundary between in-distribution discrimination and threshold transfer is itself decision-relevant knowledge. The calibration pool (adaptive refusals) and the benign pool (compliant responses) live in different regions of the fused score space. A decision system must either keep its operating context inside the calibrated region or re-calibrate online when the deployment distribution drifts toward compliant benign traffic. Third, honest boundary reporting is a decision-support feature, not a weakness: a score with a documented failure region

578 supports risk-informed routing (e.g., escalating scores that fall outside the
 579 calibrated envelope to a human reviewer) better than a threshold with an un-
 580 measured false-positive rate. These are the operational postures that follow
 581 from our measurement (Guo et al., 2024; Shafer, 1976; Xu and Li, 2025).

582 8. Limitations

583 We report the principal limitations explicitly, each with the direction we
 584 consider most promising for addressing it.

- 585 (i) **Scorer family bias.** All internal scorers are Gemma-family; cross-
 586 family anchoring bounds but does not eliminate this. Re-anchoring the
 587 protocol with at least one non-Gemma internal scorer to quantify the
 588 residual family bias directly is a natural next step.
- 589 (ii) **Chinese-language scoring.** Only the dual-judge caliber is reliable
 590 on Chinese inputs; single-scorer Chinese results are not reported. Ex-
 591 tending single-scorer Chinese reliability (e.g., Chinese-native judges or
 592 scorer fine-tuning) is future work.
- 593 (iii) **Acoustic factor confound.** A_s is implemented by voice, confounding
 594 speaker identity and gender; acoustic-attack signal is not measured at
 595 our sample window ($N = 32$). Future work will separate voice identity
 596 from prosody (e.g., TTS with matched speaker identity) and enlarge
 597 the acoustic window well beyond $N = 32$.
- 598 (iv) **Length confound.** The effect is partially—not wholly—length-
 599 mediated; within matched length bands a content effect remains,
 600 and all three audio-channel estimators are reported. Additional
 601 matched-length designs and per-band estimators remain a planned
 602 extension.
- 603 (v) **Detection transfer.** Fixed-threshold detection fails on benign inputs
 604 (75.9% FPR) and decays under adaptive attacks; only in-distribution
 605 discrimination and prompt-level robustness are claimed. Adaptive or
 606 online recalibration on deployment traffic, and an evaluation under
 607 gradient-based adversarial perturbations, are the natural next steps
 608 toward threshold transfer.

- 609 (vi) **Out-of-distribution intent.** Attack responses lack the LoRA intent
 610 signal. Future work will add a LoRA intent signal at inference time or
 611 an explicit out-of-distribution guard on the intent branch.
- 612 (vii) **Baseline coverage.** JailGuard, Cross-modal Information Check, and
 613 SALMONN-Guard were not reproducible (unreleased code/weights)
 614 and are reported as cited values only, never as measured baselines.
 615 We will re-run these baselines on release of their code or weights.
- 616 (viii) **Sequential design.** The pre-registered maximum of 500 cells/
 617 condition was exceeded by design change (v6.2→v6.5), which we
 618 disclose. A cleaner confirmatory design within the original cap is
 619 future work.
- 620 (ix) **Missing scorings.** 1.03% of P0-C responses have deterministic pars-
 621 ing failures, bounded and disclosed. Re-scoring those cells with a robust
 622 parser is a planned cleanup.

623 No fabricated values appear anywhere in this paper; every number traces
 624 to the released pipeline artifacts.

625 9. Conclusion

626 We decomposed adversarial framing against large audio-language mod-
 627 els into its semantic components and showed, under an audited paired fac-
 628 torial protocol, that narrative structure robustly increases attack success
 629 ($\Delta 27.9$ pp, OR 16.65; a larger agreement-conditioned dual-judge estimate
 630 is disclosed as inflated by selection) across languages, models, and out-of-
 631 distribution inputs. A substantial share of this effect is mediated by response
 632 elaboration, and cross-modal safety divergence is model-dependent rather
 633 than modality-intrinsic. We further introduced MSRF, a 225-parameter
 634 knowledge-driven fusion framework that discriminates framed attacks in-
 635 distribution (0.9784 ROC-AUC on the deployment frame) while exposing
 636 its own boundary conditions honestly. Narrative structure exhibits a robust
 637 causal effect under controlled prompt interventions; representing that effect
 638 as structured, multi-signal knowledge is, we argue, a more durable founda-
 639 tion for large audio-language model safety than either modal shortcuts or
 640 opaque detection.

641 **Data availability**

642 All pipeline artifacts, configurations, frozen seeds, and result files
643 are released in the anonymized companion repository accompanying this
644 manuscript; no pre-trained guard weights are redistributed.

645 **Ethics**

646 All experiments use fabricated, synthetic harmful queries (never real per-
647 sonal data), are confined to controlled research environments, and are re-
648 ported under a dual-use disclosure consistent with the study protocol and its
649 gate decisions (see the companion repository). The intent-classifier training
650 data is synthetic.

651 **CRedit authorship contribution statement**

652 **Yongjin Chen:** Conceptualization, Methodology, Software, Validation,
653 Formal analysis, Investigation, Resources, Data curation, Writing – original
654 draft, Writing – review & editing, Visualization, Supervision, Project admin-
655 istration.

656 **Songze Zhu:** Funding acquisition.

657 **Declaration of competing interest**

658 None declared.

659 **Funding**

660 This work was supported by the research project “Data Black-Industry
661 Resilience: Generation Mechanisms and Governance Countermeasures”
662 (Grant No. 25CSH113).

663 **Generative AI disclosure**

664 This manuscript was drafted with the assistance of AI writing tools for
665 language and organization; all experiments, measurements, and reported
666 numbers are human-executed and machine-verified.

667 **References**

- 668 Alanova, S., Minko, B., Sadiekh, S., Kokuykin, E., 2026. Cross-lingual jail-
669 break detection via semantic codebooks. [arXiv:2604.25716](#).
- 670 Bates, D., Mächler, M., Bolker, B., Walker, S., 2015. Fitting linear mixed-
671 effects models using lme4. *Journal of Statistical Software* 67, 1–48. doi:[10.18637/jss.v067.i01](#).
- 672
- 673 Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: A
674 practical and powerful approach to multiple testing. *Journal of the Royal*
675 *Statistical Society. Series B (Methodological)* 57, 289–300. doi:[10.1111/j.2517-6161.1995.tb02031.x](#).
- 676
- 677 Chen, G., Xia, Y., Jia, X., Li, Z., Torr, P., Gu, J., 2025. LLM jailbreak detec-
678 tion for (almost) free!, in: *Findings of the Association for Computational*
679 *Linguistics: EMNLP 2025*, pp. 5777–5807.
- 680 Cohen, J., 1968. Weighted kappa: Nominal scale agreement with provision
681 for scaled disagreement or partial credit. *Psychological Bulletin* 70, 213–
682 220. doi:[10.1037/h0026256](#).
- 683 Efron, B., Tibshirani, R.J., 1993. An Introduction to the Bootstrap. vol-
684 ume 57 of *Monographs on Statistics and Applied Probability*. Chapman
685 and Hall, New York.
- 686 Feng, B.H., Li Liang, Y.H., Liu, C.F., Chang, Y.H., Chen, Y.N., 2026a. Au-
687 dio jailbreaks in large audio-language models: Taxonomy, attack–defense
688 analysis, and cost-aware evaluation. [arXiv:2605.30031](#).
- 689 Feng, B.H., Liu, C.F., Li Liang, Y.H., Yang, C.K., Fu, S.W., Chen, Z., Lu,
690 K.H., Huang, S.F., Yang, C.H.H., Wang, Y.C.F., Chen, Y.N., Lee, H.y.,
691 2026b. Emotional damage: Investigating safety vulnerabilities of large
692 audio-language models under speaker emotional variations, in: *Proceed-*
693 *ings of the IEEE International Conference on Acoustics, Speech and Signal*
694 *Processing (ICASSP 2026)*, pp. 17792–17796. [arXiv:2510.16893](#).
- 695 Fisher, R.A., 1935. *The Design of Experiments*. Oliver and Boyd, Edinburgh.
- 696 Guo, Z., Wan, Z., Zhang, Q., Zhao, X., Zhang, Q., Kaplan, L.M., Jøsang, A.,
697 Jeong, D.H., Chen, F., Cho, J.H., 2024. A survey on uncertainty reasoning

698 and quantification in belief theory and its application to deep learning.
699 Information Fusion 101, 101987. doi:[10.1016/j.inffus.2023.101987](https://doi.org/10.1016/j.inffus.2023.101987).

700 Han, S., Rao, K., Ettinger, A., Jiang, L., Lin, B.Y., Lambert, N., Choi,
701 Y., Dziri, N., 2024. Wildguard: Open one-stop moderation tools for safety
702 risks, jailbreaks, and refusals of LLMs, in: Advances in Neural Information
703 Processing Systems 37 (NeurIPS 2024). [arXiv:2406.18495](https://arxiv.org/abs/2406.18495).

704 Inan, H., Upasani, K., Chi, J., Rungta, R., Iyer, K., Mao, Y., Tontchev,
705 M., Hu, Q., Fuller, B., Testuggine, D., Khabsa, M., 2023. Llama
706 guard: LLM-based input-output safeguard for human-ai conversations.
707 [arXiv:2312.06674](https://arxiv.org/abs/2312.06674).

708 Lakens, D., Scheel, A.M., Isager, P.M., 2018. Equivalence testing for psy-
709 chological research: A tutorial. Advances in Methods and Practices in
710 Psychological Science 1, 259–269. doi:[10.1177/2515245918770963](https://doi.org/10.1177/2515245918770963).

711 Li, H., Zhou, C., Wang, C., Liang, S., Chen, Y., Xie, Q., Ye, J., Wu, J.,
712 2026. StyleBreak: Revealing alignment vulnerabilities in large audio-
713 language models via style-aware audio jailbreak, in: Proceedings of the
714 40th AAAI Conference on Artificial Intelligence (AAAI 2026), pp. 37591–
715 37599. doi:[10.1609/aaai.v40i44.41093](https://doi.org/10.1609/aaai.v40i44.41093), [arXiv:2511.10692](https://arxiv.org/abs/2511.10692).

716 Lin, W., Li, J., Xiong, H., Liu, L., 2026. SARSteer: Safeguarding large
717 audio-language models via safe-ablated refusal steering, in: International
718 Conference on Machine Learning (ICML 2026). [arXiv:2510.17633](https://arxiv.org/abs/2510.17633).

719 Mazeika, M., Phan, L., Yin, X., Zou, A., Wang, Z., Mu, N., Sakhaee, E., Li,
720 N., Basart, S., Li, B., Forsyth, D., Hendrycks, D., 2024. HarmBench: A
721 standardized evaluation framework for automated red teaming and robust
722 refusal, in: Proceedings of the 41st International Conference on Machine
723 Learning (ICML 2024). [arXiv:2402.04249](https://arxiv.org/abs/2402.04249).

724 McNemar, Q., 1947. Note on the sampling error of the difference between
725 correlated proportions or percentages. Psychometrika 12, 153–157. doi:[10.1007/BF02295996](https://doi.org/10.1007/BF02295996).

726

727 Pan, L., Fu, Z., Zhai, Y., Tao, S., Guan, S., Huang, S., Zhang, L., Liu,
728 Z., Ding, B., Henry, F., Liu, A., Wen, L., 2025. Omni-SafetyBench: A
729 benchmark for safety evaluation of audio-visual large language models.
730 [arXiv:2508.07173](https://arxiv.org/abs/2508.07173).

731 Qian, J., Li, J., 2026. Prosody-driven jailbreaks in audio LLMs: A con-
732 trolled study and mechanistic analysis, in: Proceedings of the 34th ACM
733 International Conference on Multimedia (MM 2026). [arXiv:2607.26541](#).

734 Shafer, G., 1976. A Mathematical Theory of Evidence. Princeton University
735 Press, Princeton, NJ.

736 Shao, Z., Dou, W., Pan, Y., 2024. Dual-level deep evidential fusion: Inte-
737 grating multimodal information for enhanced reliable decision-making in
738 deep learning. Information Fusion 103, 102113. doi:[10.1016/j.inffus.2023.102113](#).

740 Shi, C., Zhang, Z., Quan, G., Pan, J., Yue, X., 2026. SRD: A model-
741 agnostic semantic framework for jailbreak mitigation in large language
742 models. Knowledge-Based Systems 350, 116560. doi:[10.1016/j.knosys.2026.116560](#).

744 Song, Z., Jiang, Q., Cui, M., Li, M., Gao, L., Zhang, Z., Xu, Z., Wang,
745 Y., Wang, C., Ouyang, G., Chen, Z., Chen, X., 2025. Audio jailbreak:
746 An open comprehensive benchmark for jailbreaking large audio-language
747 models. [arXiv:2505.15406](#).

748 Souly, A., Lu, Q., Bowen, D., Trinh, T., Hsieh, E., Pandey, S., Abbeel, P.,
749 Svegliato, J., Emmons, S., Watkins, O., Toyer, S., 2024. A strongreject
750 for empty jailbreaks, in: ICLR 2024 Workshops: Reliable and Responsible
751 Foundation Models (R2-FM). [arXiv:2402.10260](#).

752 Wei, A., Haghtalab, N., Steinhardt, J., 2023. Jailbroken: How does LLM
753 safety training fail?, in: Advances in Neural Information Processing Sys-
754 tems 36 (NeurIPS 2023). [arXiv:2307.02483](#).

755 Xie, Y., Fang, M., Pi, R., Gong, N.Z., 2024a. GradSafe: Detecting jailbreak
756 prompts for LLMs via safety-critical gradient analysis, in: Proceedings of
757 the 62nd Annual Meeting of the Association for Computational Linguistics
758 (ACL 2024). [arXiv:2402.13494](#).

759 Xie, Z., Li, B., Xu, X., Liang, Z., Yu, K., Wu, M., 2024b. Fakesound:
760 Deepfake general audio detection, in: Interspeech 2024. doi:[10.21437/Interspeech.2024-1703](#), [arXiv:2406.08052](#).

- 762 Xu, W., Li, Y., 2025. Enhancing information fusion and feature selection
763 efficiency via the PROMETHEE method for multi-source dynamic decision
764 data sets. Knowledge-Based Systems 309, 112781. doi:[10.1016/j.knosys.
765 2024.112781](https://doi.org/10.1016/j.knosys.2024.112781).
- 766 Xu, Y., Qi, X., Qin, Z., Wang, W., 2024. Cross-modality information
767 check for detecting jailbreaking in multimodal large language models, in:
768 Findings of the Association for Computational Linguistics: EMNLP 2024.
769 [arXiv:2407.21659](https://arxiv.org/abs/2407.21659).
- 770 Yi, X., Zheng, S., Wang, L., Wang, X., He, L., 2024. A safety realignment
771 framework via subspace-oriented model fusion for large language mod-
772 els. Knowledge-Based Systems 306, 112701. doi:[10.1016/j.knosys.2024.
773 112701](https://doi.org/10.1016/j.knosys.2024.112701).
- 774 Yu, Y., Jin, H., Yu, Y., Zhuang, J., Wang, H., 2026. Now you hear me: Audio
775 narrative attacks against large audio-language models, in: Proceedings
776 of the 19th Conference of the European Chapter of the Association for
777 Computational Linguistics (EACL 2026). [arXiv:2601.23255](https://arxiv.org/abs/2601.23255).
- 778 Zeng, W., Liu, Y., Mullins, R., Peran, L., Fernandez, J., Harkous, H.,
779 Narasimhan, K., Proud, D., Kumar, P., Radharapu, B., Sturman, O.,
780 Wahltinez, O., 2024a. Shieldgemma: Generative ai content moderation
781 based on gemma. [arXiv:2407.21772](https://arxiv.org/abs/2407.21772).
- 782 Zeng, Y., Lin, H., Zhang, J., Yang, D., Jia, R., Shi, W., 2024b. How johnny
783 can persuade LLMs to jailbreak them: Rethinking persuasion to challenge
784 ai safety by humanizing LLMs. [arXiv:2401.06373](https://arxiv.org/abs/2401.06373).
- 785 Zhao, X., Sun, B., Chu, X., Wu, D., Jiang, G., Zhou, H., Cai, J., 2025. A
786 decision-making approach under uncertainty based on ensemble learning
787 model with multimodal data and its application in medical diagnosis. Ex-
788 pert Systems with Applications 265, 125983. doi:[10.1016/j.eswa.2024.
789 125983](https://doi.org/10.1016/j.eswa.2024.125983).